ASSESSMENT

N+ — Network Security, Maintenance and Troubleshooting Procedures

Course: Diploma in Cybersecurity

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Section A — Concept Application

Question 1: Selecting CPU, RAM and Storage

When choosing computer parts, I would check both performance and whether the parts are supported by the motherboard.

1. CPU (Processor)

- Performance: The number of cores and the speed should match the work. Basic use needs less power than gaming, editing or other heavy tasks.
- Compatibility: The processor must fit the motherboard socket and be supported by its chipset and BIOS.

2. RAM (Memory)

- Performance: Enough RAM helps the computer run multiple programs smoothly. For many everyday systems, 8–16 GB can be a useful starting point depending on the workload.
- Compatibility: Check DDR generation, supported speed, capacity and the motherboard's recommended slots.

3. Storage

- Performance: An SSD usually starts the operating system and opens applications faster than a traditional HDD. NVMe SSDs can be faster when supported.
- Compatibility: Check the available M.2 or SATA connection, drive size and supported interface.

Checking these details before buying helps avoid compatibility problems and gives better performance.

Question 2: How Antivirus Can Slow a Computer

Antivirus software protects the computer by checking files and programs for threats. This work uses system resources.

- CPU: Scanning files and checking running programs uses processor time, so other applications may respond slowly.
- RAM: The antivirus program and its threat information use memory. If RAM is already full, the system may become less responsive.
- Storage: Real-time protection reads files when they are opened or downloaded. A full scan can also cause high disk activity.

I would check Task Manager to see which resource is heavily used. I would schedule scans at a quiet time and keep the antivirus updated. I would not turn off protection just to improve speed.

Question 3: Troubleshooting Driver or Operating-System Problems

1. Restart the computer and, if needed, start Windows in Safe Mode to check whether the issue continues with fewer drivers and services running.
2. Open Device Manager and look for devices showing warning signs.
3. If the problem started after a driver update, try Roll Back Driver. If a driver is missing or old, update it from the manufacturer's official website.
4. Run Windows Update and install suitable system updates.

5. Open Command Prompt as administrator and run sfc /scannow to check and repair protected Windows system files.
6. If the issue began recently, consider System Restore to return to a restore point from before the problem.

After troubleshooting, restart the computer and test the device or application again.

Question 4: UPS and Surge Protector

A surge protector reduces the risk of damage from a sudden voltage spike. This can help protect the computer, monitor and connected equipment.

A UPS (Uninterruptible Power Supply) provides battery power for a short time during a power cut. This gives the user a chance to save files and shut down the computer safely. A UPS may also include surge protection, depending on its model.

I would save work regularly and check the UPS battery from time to time. Neither device replaces backups of important data.

Question 5: DNS and DHCP

DNS and DHCP have different jobs but both help devices use a network.

- DHCP automatically provides settings such as an IP address, subnet mask, default gateway and DNS server.
- DNS translates a name, such as an application server's name, into an IP address.

If users can connect to a server by IP address but not by its name, I would check DNS settings and name records first. If a computer receives an address such as 169.254.x.x or cannot get normal network settings, I would investigate DHCP, the address pool and the network connection.

The symptoms should be tested before deciding which service is misconfigured.

Question 6: Firewall Blocking Legitimate Traffic

I would check the firewall carefully instead of turning it off permanently.

7. Confirm the server IP address, application port and protocol.
8. Check the workstation's IP settings and test connectivity. Ping can be used if ICMP traffic is allowed.
9. Review firewall logs for blocked connections that match the client, server or application port.
10. Use an appropriate test, such as Test-NetConnection on Windows, to check whether the required port is reachable.
11. If a rule is blocking valid traffic, add or correct a narrowly defined allow rule for the required source, destination and port.
12. Test the application again and check the logs. Do not leave the firewall disabled or allow all traffic.

This approach fixes the specific issue while keeping unnecessary network access blocked.

Section B — Practical Application

Note: Replace each figure box with your own screenshot or photo. Add only evidence from work you actually performed.

Task 1: Workstation Assembly

Aim: Assemble the main computer components and check that the system starts correctly.

13. Switch off and unplug the computer. Work on a clean surface and use anti-static precautions.
14. Place the CPU in the motherboard socket by matching the alignment mark. Close the retention mechanism without forcing it.
15. Install the CPU cooler. Apply thermal paste only if it is not already applied.
16. Insert compatible RAM in the recommended slots and make sure the clips lock.
17. Fit the M.2 SSD into its slot and secure it, or mount a SATA drive and connect its data cable.
18. Install the motherboard and power supply in the case using the correct screws and standoffs.
19. Connect the 24-pin motherboard power cable, CPU power cable, storage cables, fan headers and front-panel connectors.
20. Install the graphics card if required, secure it and connect its power cable if it needs one.
21. Connect the monitor and power cable, then start the computer.

22. Check that the system reaches the BIOS/UEFI screen and detects the CPU, memory and storage.



Figure 1: Workstation components installed and connected.



Figure 2: BIOS/UEFI screen showing detected hardware or successful POST.

Task 2: RAM Mismatch Troubleshooting

Aim: Find and correct a memory detection or compatibility problem.

23. Read any POST warning shown during startup and note the reported memory amount.
24. Shut down the computer, unplug it and check the RAM labels and motherboard manual.
25. Check whether the RAM modules are supported and whether they are installed in the recommended slots.
26. Reseat the modules carefully. Test one module at a time if the problem remains.
27. Enter BIOS/UEFI and use stable automatic memory settings. If necessary, disable XMP/EXPO while testing.
28. Save the settings, restart and check the recognised RAM amount in BIOS or Task Manager.

The system should detect the expected memory and start without the same warning. The exact cause should be recorded from the test results.



Figure 3: RAM warning or memory configuration screen, if available.



Figure 4: RAM detected correctly after troubleshooting, if available.

Task 3: LAN Topology Build

Aim: Create a small LAN in Cisco Packet Tracer with five PCs and one printer connected to a switch.

29. Add one switch, five PCs and one printer to the Packet Tracer workspace.
30. Use copper straight-through connections to connect each PC and the printer to the switch.
31. Assign each device a unique address in the same subnet. Example plan:

Device	IP address	Subnet mask
PC 1	192.168.10.2	255.255.255.0
PC 2	192.168.10.3	255.255.255.0
PC 3	192.168.10.4	255.255.255.0
PC 4	192.168.10.5	255.255.255.0
PC 5	192.168.10.6	255.255.255.0

Printer	192.168.10.7	255.255.255.0

32. Open a PC's Desktop tab, select Command Prompt and ping the printer address, for example: ping 192.168.10.7.
33. Check that the ping replies are successful. If not, check the cables, IP addresses and subnet masks.

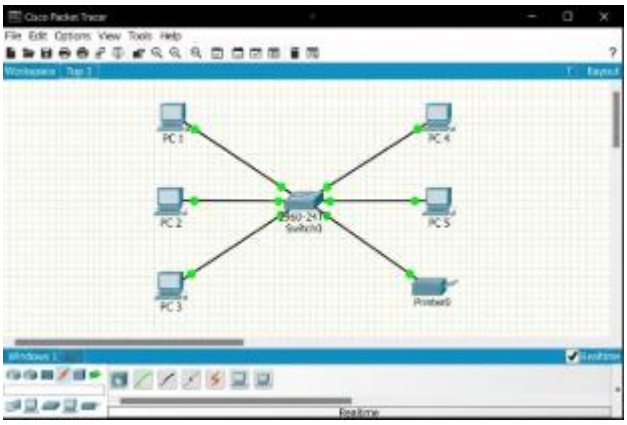


Figure 5: Packet Tracer topology showing five PCs and one printer connected to a switch.

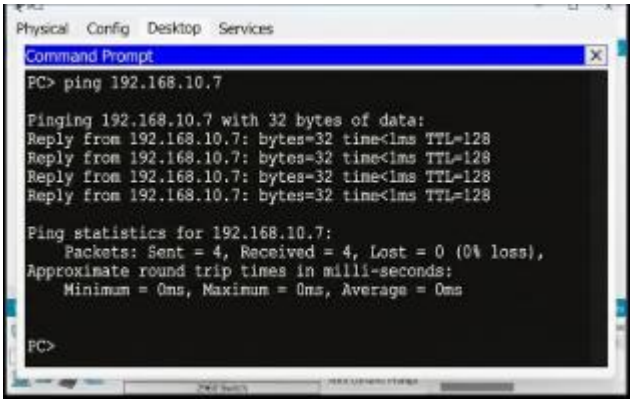


Figure 6: Successful ping test between a PC and the printer.

Task 4: Clean Operating-System Installation

Aim: Install an operating system from bootable USB media and check the startup settings.

34. Back up important files first. Create a bootable USB using the correct operating-system ISO.
35. Restart the computer and open the boot menu using the key shown by the computer manufacturer.
36. Boot from the USB and follow the installer instructions.
37. Choose the manual or custom partitioning option only after checking the target drive carefully. Deleting partitions can erase data.
38. Create or select the required OS partition(s) according to the installer and operating system being used.
39. Finish the installation, remove the USB when instructed and complete the first-time setup.
40. After reaching the desktop, install updates and required drivers. Review startup applications and disable only unnecessary ones.

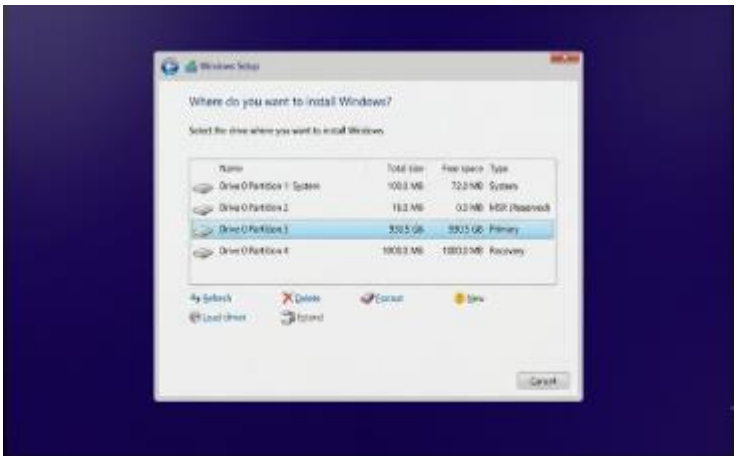


Figure 7: Operating-system installer showing the manual/custom partitioning step.

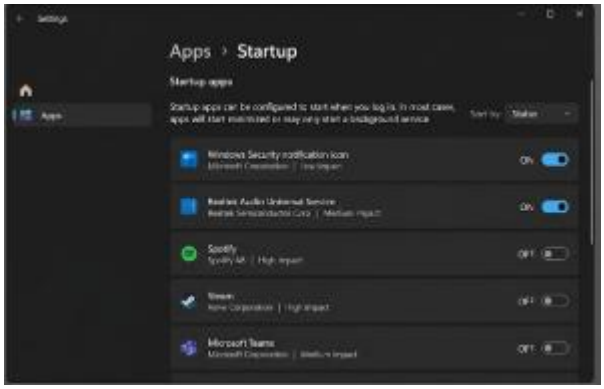


Figure 8: Installed operating system desktop or startup-app settings.

Section C — Mini Capstone Project

Scenario: Three workstations restart unexpectedly during afternoon rendering work, although they start normally in the morning.

1. Three Possible Causes

- Power supply (PSU): It may not provide stable power when the CPU and GPU are under heavy load.
- Cooling problem: Dust, a loose cooler or poor airflow may cause temperatures to become too high.
- RAM or graphics hardware: Faulty memory, an unstable GPU or an unstable setting may cause crashes under load.

2. Why the Problem May Happen in the Afternoon

Rendering creates heavy CPU and GPU use and produces heat. If the room is warmer later in the day or airflow is poor, the components may reach unsafe temperatures. A power supply or component fault may also appear only when the system is under heavy load.

3. How I Would Confirm the Cause

41. Check Windows Event Viewer or system logs for the time of each restart.
42. Monitor CPU and GPU temperatures and fan operation during a controlled workload.
43. Test memory with a suitable memory diagnostic and check the graphics driver.
44. If authorised and safe, test CPU/GPU load with suitable diagnostic tools while monitoring temperature. Stop if

temperatures become unsafe or the system behaves abnormally.

High temperatures before a shutdown may point to cooling. A sudden restart under load with normal temperatures may suggest power delivery or another hardware fault, but further tests are needed.



Figure 9: Temperature or hardware monitoring results during testing, if performed.

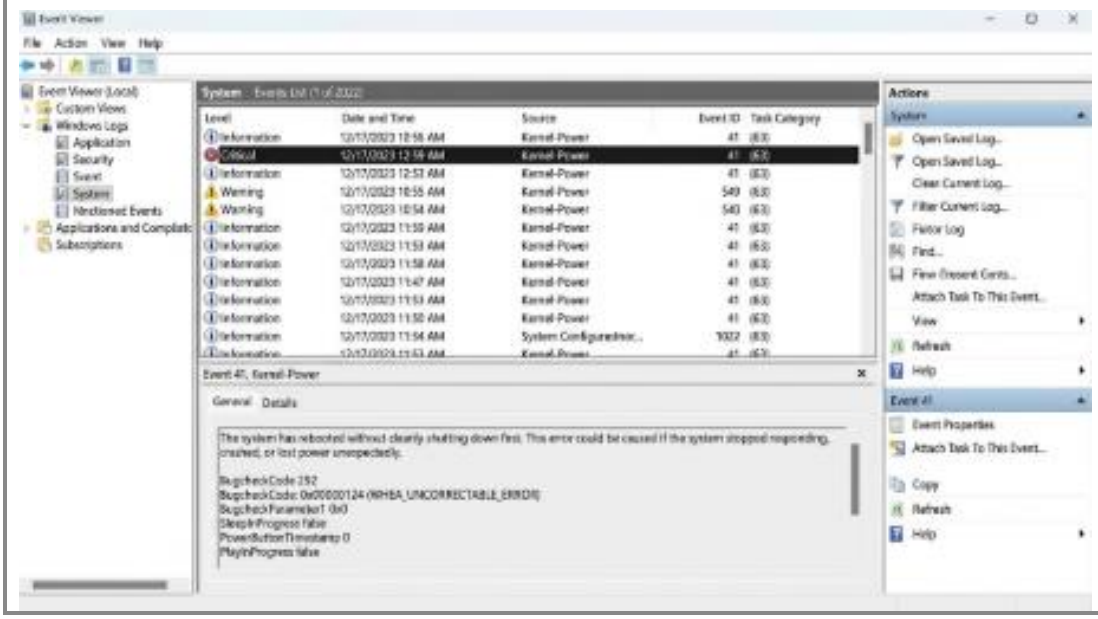




Figure 10: Event Viewer, diagnostic result or system log related to the restart, if available.

4. Three Remediation Steps

45. Clean dust from the case, heatsinks and fans; check cooler mounting and thermal paste if needed.
46. Improve airflow by checking fan direction, cable placement and case ventilation.
47. Test RAM and GPU stability, restore stable settings and update or roll back drivers if evidence points to a driver issue.

5. Is a PSU Upgrade Needed?

A PSU upgrade is needed only if testing shows that the existing unit is under-rated, faulty or unable to handle the system's power demand. If the problem is caused by overheating or RAM instability, replacing the PSU may not solve it. The cause should be confirmed before buying parts.

Section D — AI-Augmented Learning

“Make an easy IP addressing plan for a small office with five computers and one network printer. Include the subnet, subnet mask, usable host range, broadcast address, device addresses and some room for future devices. Explain the plan in simple language.”

IP Plan and Manual Check

I checked that the subnet has enough usable addresses for the devices and leaves some addresses for future use. A /28 subnet gives 16 total addresses, including the network and broadcast addresses, and 14 usable host addresses.

Item	Plan
Subnet ID	192.168.10.0/28
Subnet mask	255.255.255.240
Usable host range	192.168.10.1 – 192.168.10.14
Broadcast address	192.168.10.15
Gateway (example)	192.168.10.1
PC 1 – PC 5	192.168.10.2 – 192.168.10.6
Printer	192.168.10.7
Future addresses	192.168.10.8 – 192.168.10.14, subject to the gateway/device assignments above

Correction / Learning Note

A small /29 network has only six usable host addresses. If one address is used by the gateway, five PCs and a printer need seven addresses, so /29 is not enough. A /28 subnet provides 14 usable addresses. I also checked that the network and broadcast addresses are not assigned to a PC or printer.

The IP plan is an example for a practice network. Before using it in a real office, I would check the existing router settings and make sure the same IP addresses are not already in use.

Conclusion

In this assessment, I learned how to choose compatible computer components, troubleshoot common system and network problems, protect hardware from power issues and understand DNS, DHCP and firewall rules. I also practised the steps for assembling a workstation, checking RAM, creating a small LAN and installing an operating system. The capstone showed why it is important to check logs, temperature and hardware before replacing parts. I used AI to help draft an IP plan and then checked the subnet details myself.